A Conditional Diffusion Model With Fast Sampling Strategy for Remote Sensing Image Super-Resolution

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***Abstract*— Conventional deep learning-based methods for single remote sensing image super-resolution (SRSISR) have made remarkable progress. However, the super-resolution (SR) outputs of these methods are yet to become sufficiently satisfactory in visual quality. Recent diffusion model-based generative deep learning models are capable to enhance the visual quality of output images, but this capability is limited due to their sampling efficiency. In this article, we propose FastDiffSR, an SRSISR method based on a conditional diffusion model. Specifically, we devise a novel sampling strategy to reduce the number of sampling steps required by the diffusion model while ensuring the sampling quality. Meanwhile, the residual image is adopted to reduce computational costs, demonstrating that integrating channel attention and spatial attention begets a further improvement in the visual quality of output images. Compared to the state-of-the-art (SOTA) convolutional neural network (CNN)-based, GAN-based, and Transformer-based SR methods, our FastDiffSR improves the learned perceptual image patch similarity (LPIPS) by 0.1–0.2 and achieves better visual results in some real-world scenes. Compared with existing diffusion-based SR methods, our FastDiffSR achieves significant improvements in pixel-level evaluation metric peak signal-noise ratio (PSNR) while having smaller model parameters and obtaining better SR results on Vaihingen data with faster inference time by 2.8–28 times, showing excellent generalization ability and time efficiency. Our code will be open source at https://github.com/Meng-333/FastDiffSR.**

***Index Terms*— Conditional diffusion model, deep learning, generative models, remote sensing, super-resolution (SR).**

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I. Introduction

HIGH spatial resolution remote sensing images (RSIs) are widely used in various computer vision tasks, such as building extraction [1], water body classification [2], [3], [58], land cover mapping, and change detection [4], [59]. Limited by satellite revisit cycles and weather factors, however, continuous time series high-resolution images of the same location are not easy to obtain. With the significant development of deep learning techniques, the single-image super-resolution (SISR) method has played a convenient, efficient, and low-cost role in improving the quality of RSIs.

Over the past few years, convolutional neural networks (CNNs) have been widely used in SISR. Based on the hybrid-scale self-similarity exploitation module (HSEM), Lei and Shi [5] proposed HSENet to learn stronger feature representation. Subsequently, some Transformer-based methods continued to emerge [6], [7], [8]. Lei et al. [7] designed TransENet to better fuse multiscale low-/high-dimensional features. To further improve the visual quality of reconstructed results, some GAN-based methods [9], [10], [11] also made important contributions. Guo et al. [9] developed a novel dense generative adversarial network (NDSRGAN) that can better restore texture details. Although these methods achieve excellent performance, the visual quality of the reconstructed results still needs to be improved.

Recently, as a novel generative model, the diffusion model has gradually shown advantages in many fields [4], [12], [13], [14] and also obtained excellent performance in super-resolution (SR) tasks [15], [16], [17], [18], [19], [20]. Saharia et al. [19] proposed a method for natural image SR through iterative refinement (SR3). Subsequently, Ali et al. [15] and Liu et al. [18] introduced the SR3 model for RSI SR research. Although more realistic results were obtained than traditional GANs, the model with 2000 diffusion steps and a large number of parameters requires a long time to train and infer when generating SR images. A few months ago, Xiao et al. [20] proposed EDiffSR, a lightweight diffusion-based method to reduce model computation cost. Even so, the sampling efficiency issue is not resolved.

In response to the drawback that existing models require thousands of iterative denoising steps, we apply the diffusion model theory to single RSI SR (SRSISR) and innovatively propose a conditional diffusion model (FastDiffSR) with a

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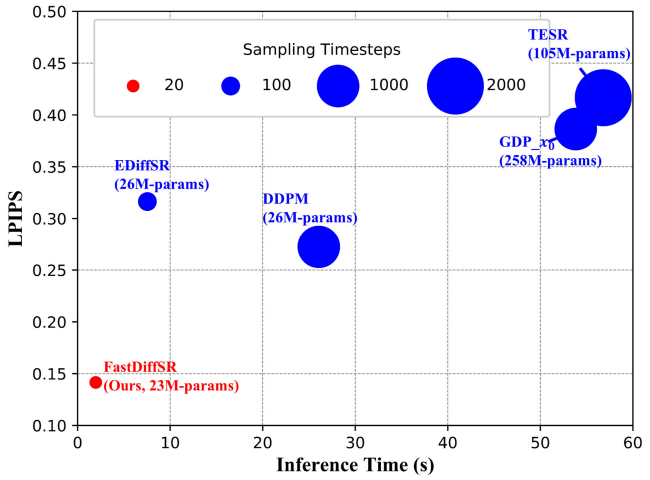


Fig. 1. Comparison of objective metrics between existing Diffusion-based SR methods on Vaihingen image for upscaling factor ×8. More quantitative results are in Section VI-E.

fast sampling strategy to improve the algorithm efficiency of the diffusion model. First, we explore a new technique that combines a linear schedule and a cosine schedule to reduce the number of diffusion steps while ensuring sampling quality. Moreover, the residual image is introduced to reduce the computational cost of the training phase, and channel attention and spatial attention are used to enhance the visual quality of common artificial objects in SR images. As shown in Fig. 1, our FastDiffSR demonstrates superior performance while being comparably faster compared to the current diffusion-based methods.

The main contributions of our article are as follows.

1) We propose a diffusion-based SR framework with sampling steps $T = 20$ (FastDiffSR) for RSIs to achieve both high visual quality and sampling efficiency.
2) We propose a novel 23M-parameter denoising network that uses the residual image as input to reduce the computational cost and improve the accuracy of noise prediction.
3) We use 22 852 images from two datasets to train our method and evaluate its performance on common artificial targets and real multispectral data. The experimental results show that our FastDiffSR method outperforms the current state-of-the-art (SOTA) diffusion-based SR methods while having a huge time efficiency advantage. There is potential for application in combination with other downstream vision tasks.

The rest of this article is arranged as follows. First, we present the related works in Section II. Section III presents the methodology of our proposed framework, fast sampling strategy, denoiser network, and training algorithm. Section IV introduces the dataset and evaluation metrics of this article. Section V provides the experimental results and Section VI presents a discussion. Section VII concludes this article.

II. Related Work

A. SR Based on Nondiffusion Models

In early times, Dong et al. [21] conducted an interesting work and proposed a three-layer convolutional network for their image SR work, which left a deep impression on people. Subsequently, neural networks gradually attracted people's attention due to their powerful nonlinear fitting capabilities. Lim et al. [22] designed an enhanced deep SR network (namely, EDSR) by increasing the network depth, and experiments have proven that it can effectively enhance the method reconstruction performance. Pan et al. [23] proposed a residual dense backprojection network (RDBPN) for RGB RSI SR. What is more, reconstruction can be accelerated by substituting the downprojection unit in the network. Dong et al. [24] designed a wide feature attention block (WAB) to improve the network representation ability. Some studies on diverse spatial scales of ground objects in RSIs and Sentinel-2 data have also been proposed [25], [26]. Although these CNN-based models have obtained good peak signal-noise ratio (PSNR) results, most of them use L1 or L2 loss functions for training, and the generated SR results are too smooth.

To generate SR images with better visual quality, many researchers have made great contributions. Ledig et al. [27] successfully reconstruct realistic textures on natural images using an adversarial training framework (SRGAN). Inspired by SRGAN, Li et al. [10] proposed an SRAGAN model for SRSISR by introducing local and global attention mechanisms to the generative adversarial network. Xu et al. [28] designed a texture enhancement generative adversarial network (TE-SRGAN) to solve blurry object edges and artifacts. Spectra-guided generative adversarial network (SpecGAN) that combines dual data source inputs of RGB image and corresponding hyperspectral image proposed by Meng et al. [11] can solve large-factor SR of RSIs. In addition, Li and Li [29] and Salgueiro Romero et al. [30] have also used generative adversarial networks to explore the spatial resolution improvement of Sentinel-2 images. GAN-based methods do achieve a better balance in pixel error and visual effects, but the instability of network training has not been solved.

Different from CNN-based and GAN-based methods, the Transformer-based method is another traditional generative model. To reduce the heavy computational cost and high GPU memory occupation of vision transformer (ViT), Liu et al. [31] designed a simple and efficient transformer (SRT) for the natural image SR task. Another efficient SR transformer (namely ESRT) work was proposed by Lu et al. [32]. Chen et al. [33] alternately apply spatial and channel self-attention in consecutive transformer blocks and propose a dual aggregation transformer (DAT) for image SR. Moreover, Transformer-based approaches have also been applied to remote sensing fields as studied in [34] and [35]. Li et al. [34] developed a multiattention SR transformer (MAST) to improve the model's utilization of multiscale information and perception ability of local detailed features. They perform well on image SR tasks but require large amounts of data and computing resources for pretraining.

B. SR Based on Conditional Diffusion Models

Owing to the excellent performance of the diffusion model in image generation tasks, some researchers have started researching diffusion model methods for their image SR.

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Ho et al. [17] first proposed a denoising diffusion probabilistic model (DDPM) to obtain high-quality natural image synthesis results. Subsequently, a series of improved methods were proposed [36], [37], [38], [39], [40], [41], [42]. Li et al. [43] proposed a novel conditional diffusion model (SRDiff) guided by LR input to achieve realistic single-image SR prediction. The introduced residual prediction can accelerate model convergence. Ho et al. [44] also developed a cascade diffusion model that gradually increases image resolution to improve sample quality. In [19], by applying multistep iteration ideas in DDPM, the authors present a new method to image SR through repeated refinement (SR3), which achieves stronger photorealistic outputs than the GAN-based method in the face image $\times 8$ SR task. More recently, Peebles and Xie [45] explored a novel class of diffusion models by designing a transformer architecture and replacing the commonly used U-Net backbone. The proposed transformer network also has good scalability. Shang et al. [46] proposed a new diffusion probability model (namely, ResDiff) based on a residual structure to achieve the SISR task by combining a CNN and a DPM. Although they have excellent performance on natural images, they are challenging for RSIs with rich spectral and shape features.

Research on SRSISR combined with a diffusion model is still in its infancy. The study proposed by Liu et al. [18] is one of the first diffusion-based studies applied to the SRSISR task. To better recover high-resolution images, Han et al. [47] developed a new method with an efficient hybrid conditional diffusion model (EHC-DMSR). Among them, the conditions for guiding image generation are extracted through the transformer network and CNN. In addition, a Fourier high-frequency spatial constraint is designed to constrain the diffusion model and generate more high-frequency information. Similar to the EHC-DMSR work, Ali et al. [15] proposed a novel algorithm named TESR to artificially improve the resolution of RS images. Xiao et al. [20] designed an efficient DPM (EDiffSR), which includes a conditional prior enhancement module (CPEM) and an efficient activation network (EANet). The proposed CPEM module can help EDiffSR extract enriched conditions, and the EANet network can reduce the computational budget while having favorable noise prediction performance. There is also a diffusion-based approach (namely HSR-Diff) that has been applied to hyperspectral image SR, as explored in [48]. However, most existing diffusion-based RSI SR approaches focus on designing new condition guidance modules and improving noise predictors but ignore the most important problem of sampling efficiency, which limits the practical application of the model.

III. Methodology

As a class of probabilistic generative models, diffusion models aim to generate high-quality data for a given training data input by adding T-timestep noise at different scales to destroy the sample structure distribution and learn to reverse a process that gradually removes the noise. In this work, we propose an SR approach, FastDiffSR, based on the diffusion model to reconstruct realistic texture details of RSIs.

A. Diffusion Framework

A diffusion model includes a forward diffusion process and a reverse inference process. Fig. 2 shows the architecture of the proposed diffusion framework, and then, we will introduce the concept of two processes in our FastDiffSR, the fast sampling module, the denoiser network, and the corresponding training algorithm in detail.

1) Forward Diffusion Process: Stage 1 of the proposed FastDiffSR method is named the forward diffusion process, which gradually corrupts the original data by applying Gaussian noise. Let x_0 be the original data (uncorrupted) and x_t be the result data (corrupted) after adding noise to the original data. According to the Markov process, we have

$$p(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t \cdot \mathbf{I}\right) \quad t \in \{1, \ldots, T\} \tag{1}$$

$$p(x_1, \ldots, x_T \mid x_0) = \prod_{t=1}^{T} p(x_t \mid x_{t-1}), \quad t \in \{1, \ldots, T\} \tag{2}$$

where $\mathcal{N}$ is the normal distribution, T is diffusion timesteps, $p(x)$ represents the variable distribution, and β_t is a constant and $\in[0, 1)$.

Our formula (1) can be further understood as

$$x_t = \sqrt{1-\beta_t} \cdot x_{t-1} + \sqrt{\beta_t} \cdot z_t \tag{3}$$

where $z_t \in \mathcal{N}(\mathbf{0}, \mathbf{I})$. Setting $\alpha_t = 1 - \beta_t$, x_{t-1} can be defined as

$$x_{t-1} = \sqrt{\alpha}_{t-1} \cdot x_{t-2} + \sqrt{1-\alpha}_{t-1} \cdot z_{t-1}. \tag{4}$$

Given original data x_0 and variance schedule $\beta_1, \ldots, \beta_T$, combining formulas (3) and (4), we can directly get arbitrary x_t from x_0

$$x_t = \sqrt{\bar{\alpha}_t} \cdot x_0 + \sqrt{1-\bar{\alpha}_t} \cdot \epsilon \tag{5}$$

where $\bar{\alpha}_t$ is $\prod_{i=1}^{t} \alpha_i$, $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

When $t = T$, if $\bar{\alpha}_t \to 0$, then, according to (5), we have x_T that is approximately equal to the standard normal distribution ϵ.

2) Reverse Inference Process: Stage 2 of the proposed FastDiffSR method is named the reverse inference process, and it is defined by a Markov chain with learned Gaussian transitions starting at $q(x_T) = \mathcal{N}(x_T; \mathbf{0}, \mathbf{I})$

$$q(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu(x_t, t), \sigma(x_t, t)^2 \cdot \mathbf{I}) \tag{6}$$

$$q\left(x_0, \ldots, x_{T-1} \mid x_T\right) = \prod_{t=1}^{T} q(x_{t-1} | x_t). \tag{7}$$

In the inference phase, its goal is to train a neural network to predict the noise distribution added in the forward phase and then slowly estimate the original data x_0 by removing the noise. According to the Bayes rule and Gaussian distribution, we have

$$q(x_{t-1} \mid x_t) = \frac{q(x_t | x_{t-1}) \cdot q(x_{t-1})}{q(x_t)} \tag{8}$$

$$\mathcal{N}\left(\mu, \sigma^2\right) \propto \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{9}$$

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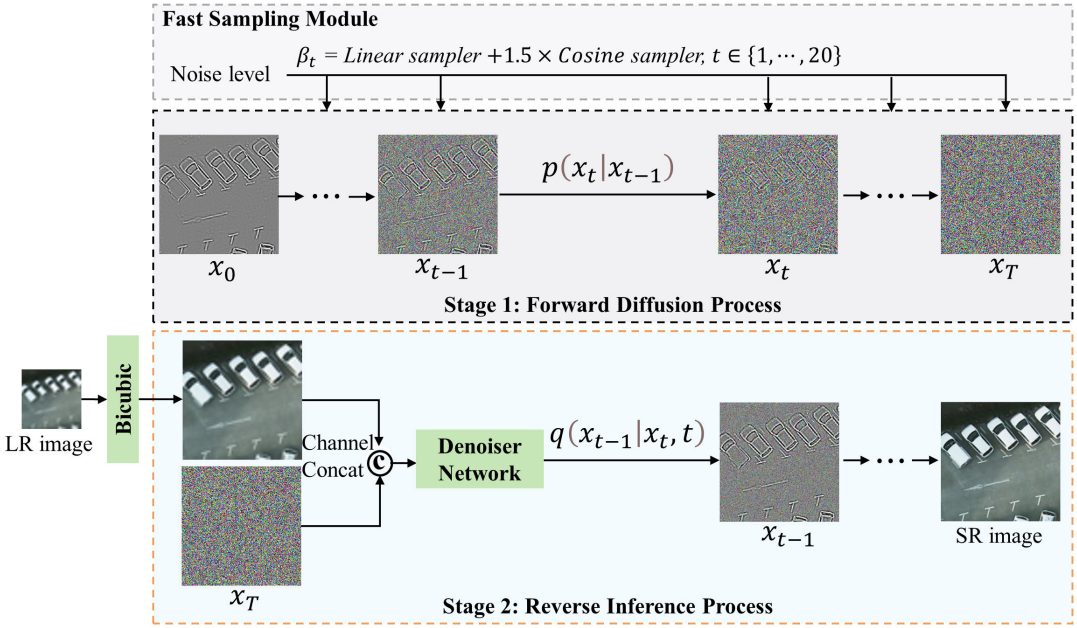


Fig. 2. Overview structure of the proposed FastDiffSR diffusion framework. The two stages are noise addition and denoising according to the fast sampling module with $T = 20$ steps. x_0 is the residual image of the subtraction between the HR image and the bicubic image.

Therefore, from (3) and (5), we get

$$q(x_t \mid x_{t-1}) \sim \mathcal{N}(\sqrt{\alpha_t} \cdot x_{t-1}, (1-\alpha_t) \cdot \mathbf{I}) \tag{10}$$

$$q(x_t) \sim \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot x_0, (1-\bar{\alpha}_t) \cdot \mathbf{I}) \tag{11}$$

$$q(x_{t-1}) \sim \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}} \cdot x_0, (1-\bar{\alpha}_{t-1}) \cdot \mathbf{I}) \tag{12}$$

$$q(x_{t-1} \mid x_t) \propto \exp\left(-\frac{1}{2} \cdot \left\{ \frac{\left(x_t - \sqrt{\alpha_t} x_{t-1}\right)^2}{1-\alpha_t} + \frac{\left(x_{t-1} - \sqrt{\bar{\alpha}_{t-1}} x_0\right)^2}{1-\bar{\alpha}_{t-1}} - \frac{\left(x_t - \sqrt{\bar{\alpha}_t} x_0\right)^2}{1-\bar{\alpha}_t} \right\}\right). \tag{13}$$

Simplifying (13), we define

$$q(x_{t-1} \mid x_t) \propto \exp\left(-\frac{1}{2} \cdot \left(A \cdot x_{t-1}^2 + B \cdot x_{t-1} + C\right)\right) \tag{14}$$

where A is $\frac{1-\bar{\alpha}_t}{\beta_t(1-\bar{\alpha}_{t-1})}$, B is $-2(\frac{\sqrt{\alpha_t} x_t}{\beta_t} + \frac{\sqrt{\bar{\alpha}_{t-1}} x_0}{1-\bar{\alpha}_{t-1}})$, and C is an unimportant constant. From (5), x_0 is defined as

$$x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t} \cdot \epsilon\right). \tag{15}$$

Then, we conclude that the mean $\mu(x_t, t)$ and variance $\sigma(x_t, t)$ are

$$\mu(x_t, t) = -\frac{B}{2A} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \cdot \tilde{\epsilon}\right) \tag{16}$$

$$\sigma(x_t, t)^2 = \frac{1}{A} = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \tag{17}$$

According to (16) and (17), the inference process (6) can be expressed as

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \cdot \tilde{\epsilon}\right) + \sqrt{\tilde{\beta}_t} \cdot z \tag{18}$$

where $t \in \{T, T-1, \ldots, 1\}$, $\tilde{\epsilon}$ is the noise predicted by the neural network, $z \in \mathcal{N}(\mathbf{0}, \mathbf{I})$, α_t and β_t are constant hyperparameters, and $\tilde{\beta}_t$ is approximately equal to β_t.

B. Fast Sampling Module

One of the notable advantages of our FastDiffSR is that it can achieve fast sampling, which does not require hundreds or even thousands of sampling steps while maintaining better sampling quality. As shown in Fig. 3, we designed a new sampling technique with fewer steps to achieve fast sampling of the conditional diffusion model. Specifically, our sampling strategy contains a linear schedule and a cosine schedule to ensure that the image will not be completely corrupted quickly before $T/2$ steps and finally let $\bar{\alpha}_t$ converge to zero when $T = 20$. We define our β

$$\beta = \beta_{\text{linear}} + 1.5 \times \beta_{\text{cosine}}. \tag{19}$$

In our experiments, we set $T = 20$ and constant β_{linear} increases linearly from $\beta_1 = 1\text{e}^{-6}$ to $\beta_T = 1\text{e}^{-2}$. For β_{cosine}, if the coefficient is too small, it will not converge to 0 when $t = T$; if it is too large, it will be discontinuous and converge prematurely. Therefore, we set the scaling coefficient to 1.5, and then, we follow [39] to clip β to be no larger than 0.999. The results of different sampling strategies are discussed in Section VI-C.

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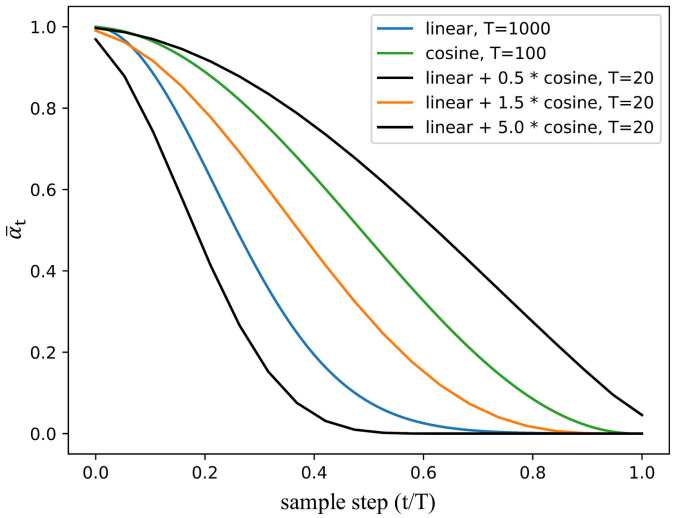


Fig. 3. $\bar{\alpha}_t$ throughout diffusion in the three sample schedules. The total T of linear and cosine sampling methods are 1000 and 100, respectively, and our number of steps T is 20.

C. Denoiser Network and Training-Inference Algorithm for FastDiffSR

1) Denoiser Network: An important component of our proposed FastDiffSR model is the denoiser network. As depicted in Fig. 4, a U-Net network with temporal encoding is adopted to predict each step of noise added in the forward diffusion process. Some studies, such as SR3 [19] and TESR [15], directly predict HR images, which we have not adopted. Instead, we adjust x_0 of (5) to be the residual value of x_{HR} image and Bicubic(x_{LR}) image to reduce the computational burden.

Specifically, we take the latent variable x_t, Bicubic(x_{LR}) and time t as inputs, and output is the predicted noise $\tilde{\epsilon}$. In Fig. 4, our U-Net network contains four DownBlocks (DBs), one MiddleBlock (MB), and four UpBlocks (UBs). The DB consists of two residual blocks and one downsampling layer. Each residual block contains two convolutional blocks that consist of a swish activation layer and a convolution layer. The MiddleBlock consists of two residual blocks and two attention modules, which are added in between. Benefiting from the work of SRAGAN, a channel local attention module (CLAM) and a spatial local attention module (SLAM) are introduced to us. Similar to DB, UB consists of two residual blocks and one upsampling layer. Finally, a convolutional block is used to generate $\tilde{\epsilon}$ as the predicted noise at timestep $t-1$, and then, x_{t-1} is recovered according to (18). Their formulation can be written as

$$f_{\text{conv block}}(x) = f_{\text{conv}}^{3\times3}(f_{\text{swish}}(x)) \tag{20}$$

$$f_{\text{res block}}(x) = f_{\text{conv block}}(f_{\text{conv block}}(x)) \tag{21}$$

$$f_{\text{DB}}(x) = f_{\text{downsample}}(f_{\text{res block}}(f_{\text{res block}}(x))) \tag{22}$$

$$f_{\text{MB}}(x) = f_{\text{res block}}(f_{\text{SLAM}}(f_{\text{CLAM}}(f_{\text{res block}}(x)))) \tag{23}$$

$$f_{\text{UB}}(x) = f_{\text{upsample}}(f_{\text{res block}}(f_{\text{res block}}(x))) \tag{24}$$

$$\tilde{\epsilon} = f_{\text{conv block}}(f_{\text{UB}}(\ldots(f_{\text{MB}}(f_{\text{DB}}(\ldots(f_{\text{DB}}(x))))))) \tag{25}$$

Algorithm 1 Training Algorithm for FastDiffSR

1: **Input**: Dataset $D = \left\{(Bicubic(x_{LR}^n), x_{HR}^n)\right\}_{n=1}^{N}$, diffusion timesteps T, constants β_1 and β_T
2: Initialize denoiser network $\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
3: **repeat**
4: Sample $(Bicubic(x_{LR}^n), x_{HR}^n)$
5: Sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $t \sim$ Uniform($\{1, \ldots, T\}$)
6: Compute $x_t = \sqrt{\bar{\alpha}_t} \cdot (x_{HR} - Bicubic(x_{LR})) + \sqrt{1-\bar{\alpha}_t} \cdot \epsilon$
7: Perform the gradient step
8: $\nabla_\theta \parallel \epsilon - \tilde{\epsilon}(x_t, Bicubic(x_{LR}), t) \parallel$
9: **until** converged

Algorithm 2 Inference Algorithm for FastDiffSR

1: **Input**: $Bicubic(x_{LR})$, diffusion timesteps T
2: Sample $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
3: **for** $t = T, T-1, \ldots, 1$ **do**
4: Sample $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $z = 0$
5: Compute $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \cdot \tilde{\epsilon}(x_t, Bicubic(x_{LR}), t)\right) + \sqrt{\tilde{\beta}_t} \cdot z$
6: **end for**
7: return $x_0 + Bicubic(x_{LR})$ as SR result

where x represents the network input and $\tilde{\epsilon}$ is the predicted noise.

2) Training Algorithm: Our training algorithm is shown in Algorithm 1, the dataset $D = \{(\text{Bicubic}(x_{\text{LR}}^n), x_{\text{HR}}^n)\}_{n=1}^{N}$ is used to train FastDiffSR with the diffusion timesteps T and constants β_1 and β_T (Line 1). We randomly initialize the denoiser network $\tilde{\epsilon}$ that obeys the standard normal distribution (Line 2). Then, we randomly sample images from the dataset D (Line 4) and sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and t from the collection $\{1, \ldots, T\}$ (Line 5). Next, the latent variable x_t is computed (Line 6) and the denoiser network is optimized with a gradient step (Line 8).

3) Inference Algorithm: As shown in Algorithm 2, we let our U-Net network accept Bicubic(x_{LR}) image as the conditioning input during T-timesteps FastDiffSR inference process (Line 1). When $t = T$, the latent variable x_T is sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ (Line 2) and the iterative procedure starts (Line 3). If $t > 1$, we sample $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ (Line 4) and calculate x_{t-1} using the U-Net network $\tilde{\epsilon}$ with x_t, Bicubic(x_{LR}), and t as inputs (Line 5). If $t = 1$, we make $z = 0$ (Line 4) and compute the final x_0 (Line 5). Then, the SR result is obtained by adding the residual image x_0 on the Bicubic(x_{LR}) image (Line 7). Fig. 5. visualizes some images at different steps.

IV. Datasets and Evaluation Metrics

A. Datasets

In our experiments, three datasets are used to evaluate the effectiveness of our method, and Fig. 6 provides some example images. The first dataset is Potsdam from the ISPRS 2-D semantic labeling contest. It was captured over Potsdam in Germany and contains 38 images. The dataset is manually labeled into six classes, including buildings, background, cars, impervious surfaces, low vegetation, and trees. These images

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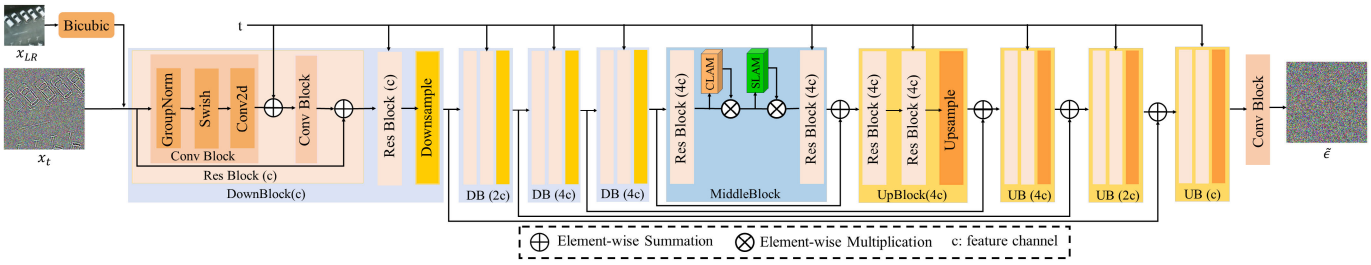


Fig. 4. Architecture of the denoiser network in our FastDiffSR.

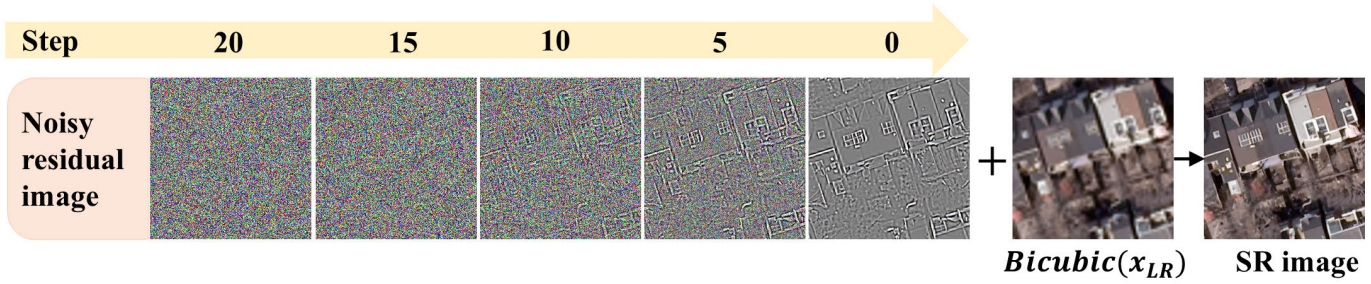


Fig. 5. Visualization of images at different steps in the inference process.



(a)



(b)

Fig. 6. Example images from the Potsdam and Toronto datasets. (a) "4_11.tif" and "7_7.tif," Potsdam. (b) "03757.tif," Toronto.

have the same size of 6000 × 6000 pixels with a ground resolution of 5 cm.

The second is the Toronto dataset [49], an urban classification and 3-D reconstruction dataset provided by ISPRS WG III/4. The dataset contains representative scene characteristics of a modern megacity in North America, including a mixture of low-/high-story buildings with a wide variety of rooftop structures and street and road features. It has 13 images of 11 500 × 7500 pixels with three channels (R-G-B) at a 0.15 m GSD, including cars, parking lots, roads, street furniture, and trees.

The third UC Merced data [50] are only used for real-world analysis without downsampling. To save the inference cost, we randomly selected seven images of four scenes from UC Merced and cropped them to 128 × 128 as model input.

In the Potsdam dataset, we set 2_14.tif, 3_14.tif, 4_14.tif, 4_15.tif, 5_14.tif, 5_15.tif, 6_14.tif, and 6_15.tif as test data (a total of 3272 patches) and the remaining for training (total 12 321 patches). Regarding the Toronto dataset, we selected 03553.tif, 03747.tif, 03753.tif, 03757.tif, 03945.tif, and 03949.tif for model training (total 5983 patches) and 03559.tif for testing (total 1276 patches). Considering the GPU memory capacity, we crop images to a size of 256 × 256 pixels (with no overlap). When scale factors are 4 and 8, their corresponding LR images are generated by bicubic interpolation, which are 64 × 64 pixels and 32 × 32 pixels, respectively.

B. Evaluation Metrics

Three commonly used reference-based evaluation metrics are adopted by us to objectively evaluate the effectiveness of different methods. They are PSNR [51], frechet inception distance (FID) [52], and learned perceptual image patch similarity (LPIPS) [53]. Lower FID and LPIPS values and higher PSNR values imply better SR quality.

For real-world SR results without ground-truth images as references, we also adopt no-reference image evaluation metrics recently used in some RSI SR studies [20], [54], such as entropy [55] and natural image quality evaluator (NIQE) [56]. A higher entropy value and smaller NIQE value imply better SR quality.

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C. Implementation Details

Our study obtained 22 852 high-resolution RSIs (256 × 256 size), of which 18 304 images were for training and 4548 images were for testing. All methods were implemented in PyTorch and trained on an NVIDIA A100-SXM4-40-GB GPU. Our FastDiffSR method uses the Adam [57] algorithm with betas as the default value and a learning rate of 0.0002. The training process (200 epochs) takes approximately 48 h with a batch size of 4, timesteps $T = 20$, and the loss function is L1 loss.

V. Experimental Results

In this article, we quantitatively compare the proposed method with the baseline bicubic method and nine excellent SR models, including SwinIR [8], HSENet [5], TransENet [7], NDSRGAN [9], HAT [6], DDPM [17], TESR [15], GDP_x_0 [16], and EDiffSR [20]. HSENet, TransENet, and EDiffSR are networks for RSI SISR published in IEEE Transactions on Geoscience and Remote Sensing. NDSRGAN is a GAN-based network for SISR of RSIs. SwinIR and HAT are Transformer-based networks for SISR of natural images published at the International Conference on Computer Vision 2021 and IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2023. DDPM and GDP_x_0 are diffusion-based networks for natural images published in 2020 and CVPR 2023. TESR is also diffusion-based network published in Remote Sensing in 2023 for SRSISR.

The hyperparameters of all baseline methods are the same as in the original paper and have been retrained on our dataset. It takes about two days for the nondiffusion-based methods to train 100 epochs, and about 4–6 days for the diffusion-based methods to train 200–400 epochs.

A. Results on the Potsdam Dataset

In this experiment, Table I lists the results of our FastDiffSR method and nine comparison models with ×4 and ×8 scales on the Potsdam dataset. For the ×4 scale task, the PSNR result of FastDiffSR is 0.143 dB higher than the next best HAT and achieves the best FID. Due to thousands of sampling steps, DDPM achieves better LPIPS but performs poorly on the PSNR metric. All model's parameters are also listed in Table I, and our FastDiffSR model size is lower than TransENet, TESR, and GDP_x_0, but it gets better performance in the ×4 scale task, which implies that our FastDiffSR has an efficiency advantage. For the eight-time task, the proposed FastDiffSR has significant advantages over CNN- and Transformer-based methods (such as HSENet, SwinIR, TransENet, and HAT) in visual quality, 0.23883–0.26825 lower in LPIPS, while PSNR metric is inferior to them. Compared with diffusion-based (such as DDPM, TESR, GDP_x_0, and EDiffSR) methods, our FastDiffSR has huge improvements in PSNR value and obtains the next best LPIPS result.

Fig. 7 shows ×4 and ×8 visual comparisons with ten SR methods. In Fig. 7(a), the bicubic method has the worst results. The results of SwinIR, HSENet, TransENet, and HAT are too smooth, and the visual perception is far from the ground-truth image. GAN-based NDSRGAN achieved slightly better LPIPS but was still lower than the diffusion-based method. In addition, both EDiffSR and GDP_x_0 methods have blur problems. The DDPM and the TESR methods seem to better preserve the spatial details of the HR image, but our FastDiffSR still obtains the best performance. From Fig. 7(b), our FastDiffSR also obtains the best in all of PSNR and LPIPS and is closer to the ground-truth image than other approaches.

TABLE I
Quantitative Results of Various SR Approaches on the Potsdam Dataset at ×4 and ×8 Scales

Method	x4			
	PSNR	FID	LPIPS	Params
Bicubic	30.689	63.785	0.4432	/
SwinIR	32.919	44.590	0.2999	11.852M
HSENet	32.932	44.192	0.3014	5.433M
TransENet	32.768	41.310	0.2878	37.459M
NDSRGAN	30.529	51.209	0.2137	17.510M
HAT	<u>32.987</u>	44.932	0.2962	25.821M
DDPM	31.989	<u>24.307</u>	**0.0997**	26.166M
TESR	32.074	28.770	0.1209	105.128M
GDP_x_0	32.324	39.232	0.1187	258.844M
EDiffSR	30.860	36.669	<u>0.1104</u>	26.790M
Ours	**33.130**	**22.832**	0.1152	23.229M
Method	x8			
	PSNR	FID	LPIPS	Params
Bicubic	28.131	138.798	0.6420	/
SwinIR	29.547	113.589	0.4827	11.999M
HSENet	**29.561**	114.326	0.4871	5.581M
TransENet	29.533	103.902	0.4577	37.607M
NDSRGAN	25.694	132.623	0.4280	17.510M
HAT	<u>29.558</u>	112.549	0.4666	25.821M
DDPM	28.744	80.461	0.2359	26.166M
TESR	28.664	<u>76.565</u>	**0.2178**	105.128M
GDP_x_0	27.852	94.312	0.2287	258.844M
EDiffSR	27.600	79.043	0.2243	26.790M
Ours	29.393	**52.136**	<u>0.2188</u>	23.229M

B. Results on the Toronto Dataset

In this experiment, Table II gives the results of our FastDiffSR method and nine comparison models with ×4 and ×8 scales on the Toronto dataset. For the ×4 scale task, FastDiffSR achieved excellent performance. Compared with the next best DDPM model, our FastDiffSR is +1.008 dB in PSNR and −0.0011 in LPIPS. For the ×8 scale task, our FastDiffSR achieves the best FID and LPIPS results and acceptable PSNR results. Specifically, compared with the HSENet model, the LPIPS metric has improved by more than 96.6%. Compared with the EDiffSR model, 1.343 dB is higher in PSNR, 39.261 is lower in FID, and 0.0157 is lower in LPIPS. Overall, our FastDiffSR achieves more satisfactory results on the ×8 scale task than on the ×4 scale task and outperforms the Diffusion-based methods in all metrics.

Fig. 8 also lists ×4 and ×8 visual comparisons with different SR methods. From Fig. 8(a), although SwinIR, HSENet, TransENet, and HAT can obtain decent PSNR values, they cannot recover accurate building window details in some challenging cases. DDPM, EDiffSR, and GDP_x_0 can only restore part of the building window. In comparison, FastDiffSR can

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Fig. 7. Visual comparisons of SR results with different SOTA methods on the Potsdam dataset. (a) ×4 factor. (b) ×8 factor.

TABLE II

QUANTITATIVE RESULTS OF VARIOUS SR APPROACHES ON THE TORONTO DATASET AT ×4 AND ×8 SCALES. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Method	x4			
	PSNR	FID	LPIPS	Params
Bicubic	24.722	107.589	0.4691	/
SwinIR	26.946	77.645	0.2285	11.852M
HSENet	<u>27.555</u>	74.868	0.2319	5.433M
TransENet	26.884	62.745	0.2260	37.459M
NDSRGAN	25.653	44.778	0.1584	17.510M
HAT	27.545	78.677	0.2214	25.821M
DDPM	26.813	**38.377**	0.1022	26.166M
TESR	26.983	<u>41.259</u>	0.1216	105.128M
GDP_x_0	26.958	51.146	0.1251	258.844M
EDiffSR	26.170	54.313	**0.0985**	26.790M
Ours	**27.901**	42.273	<u>0.1011</u>	23.229M
Method	x8			
	PSNR	FID	LPIPS	Params
Bicubic	21.738	164.404	0.7247	/
SwinIR	23.017	155.212	0.4056	11.999M
HSENet	**23.270**	162.713	0.4148	5.581M
TransENet	<u>23.229</u>	133.171	0.3963	37.607M
NDSRGAN	20.821	151.065	0.3484	17.510M
HAT	23.191	159.423	0.3699	25.821M
DDPM	22.652	94.789	0.2496	26.166M
TESR	22.689	<u>74.490</u>	0.2285	105.128M
GDP_x_0	22.280	136.622	0.2783	258.844M
EDiffSR	21.810	108.510	<u>0.2267</u>	26.790M
Ours	23.153	**69.249**	**0.2110**	23.229M

recover more accurate structures. We can see from Fig. 8(b) that bicubic interpolation cannot correctly reconstruct the HR image under eight-time downsampling. Although TransENet obtains the best PSNR result, the image visual quality scores are as blurry as SwinIR and HSENet. The NDSRGAN method generates very bad pixel values, even lower than the baseline bicubic. In addition, DDPM, EDiffSR, TEST, and GDP_x_0 reconstructed incorrect car textures. HAT does not have a good balance between image pixel value and visual quality. These visual comparisons demonstrate that our FastDiffSR is still able to reconstruct a more accurate ground-truth image even at an eight-time scale factor.

C. Evaluation of Real-World Multispectral Images

To further evaluate the SR performance of our FastDiffSR, we conducted inference experiments (4x SR task) on real-world RSIs without downsampling. A total of seven images (128 × 128) of intersection, mediumresidential, parkinglot, and tenniscourt scenes from the UC Merced dataset were used as input data. Considering that there is no ground truth, some common no-reference image evaluation metrics entropy and NIQE in RSI SR research are employed by us for quantitative comparison of SR results. A higher entropy value and smaller NIQE value imply better SR quality.

Table III shows the entropy and NIQE results of different SOTA methods on the UC Merced test set. It can be seen that our FastDiffSR achieves the best NIQE and the next best entropy, which is slightly inferior to EDiffSR. Fig. 9 also displays the visualization results of different methods on mediumresidential and tenniscourt scenes. As can be seen, most diffusion-based methods can generate satisfactory results, and the locations of the roof and swimming pool in the mediumresidential47 image are closer to human visual perception.

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Fig. 8. Visual comparisons of SR results with different SOTA methods on the Toronto dataset. (a) ×4 factor. (b) ×8 factor.

From the tenniscourt15 image, the bicubic and TESR methods have the worst results. Although DDPM and EDiffSR obtain clear car details, other regions they generate are blurrier than nondiffusion-based methods, such as parking lines and tree crowns. In contrast, only our FastDiffSR can obtain both accurate and realistic SR results. These experimental results show that our proposed method also performs well in some real-world scenes.

VI. Discussion

In this section, we conduct a series of comparative experiments on the Toronto dataset to explore the impact of U-Net network depth, two attention mechanisms, different sample schedules, our fast sampling module, and the learning rate parameter on SR performance. All experiments were conducted on an ×8 scale task.

A. Depths of U-Net Network

Different network depths can significantly affect the parameters and SR performance of our FastDiffSR model. In this section, we compare the U-Net network with different numbers of DBs and UBs to explore the impact of network depth. The metric results and model parameters of five groups of experiments are given in Table IV. Experiments imply that increasing network depth can significantly improve the model performance on all three metrics to the best and then decline. The 23.229M-parameter FastDiffSR model obtained the best results except for the FID metric, and the improvements over the model with two DBs and UBs reached 0.329 dB in PSNR, 42.72 in FID, and 0.0447 in LPIPS value.

B. Effectiveness of Attention Mechanism

To determine the importance of the attention mechanism in our U-Net network, taking the MiddleBlock without attention module as the baseline, we set channel attention, spatial attention, and channel + spatial attention and conducted ablation experiments to verify their SR performance. The experimental results of FastDiffSR of adding three attention modules, including only CLAM, only SLAM, and CLAM + SLAM, are given in Table V. The results show that the model using attention modules can improve the visual quality score LPIPS. Specifically, when the attention mechanism is applied to the model, the PSNR metric will decrease slightly, but −47.009 in FID and visual quality LPIPS can be improved by 12%.

C. Discussion of Different Sample Schedules

For diffusion-based methods, the hyperparameter diffusion timestep is crucial to the model inference time and

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TABLE III

ENTROPY AND NIQE RESULTS OF DIFFERENT SOTA METHODS ON THE UC MERCED TEST SET. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Metric	Bicubic	SwinIR	HSENet	TransENet	NDSRGAN	HAT	DDPM	TESR	GDP_x_0	EDiffSR	Ours
entropy	7.0656	7.1032	7.0826	7.0853	7.0528	7.1177	7.1204	7.0682	7.0848	**7.1838**	<u>7.1254</u>
NIQE	22.4352	19.6321	19.9131	20.319	19.2143	20.3849	<u>16.5316</u>	18.8250	19.2593	17.6413	**16.3418**



Fig. 9. Visual comparison of ten methods for 4x SR on the UC Merced images. Here, we show two scenes.

TABLE IV

EXPERIMENTAL RESULTS OF FASTDIFFSR WITH DIFFERENT PARAMETERS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Depth	PSNR	FID	LPIPS	Params
[1, 2]	22.824	111.969	0.2557	3.820M
[1, 2, 4]	<u>22.914</u>	93.676	0.2458	14.006M
[1, 2, 4, 4]	**23.153**	<u>69.249</u>	**0.2110**	23.229M
[1, 2, 4, 8]	22.818	**66.731**	<u>0.2207</u>	54.534M
[1, 2, 4, 8, 8]	22.896	87.101	0.2375	91.215M

TABLE V

EXPERIMENTAL RESULTS OF FASTDIFFSR WITH DIFFERENT ATTENTION BLOCKS. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Attention module	PSNR	FID	LPIPS	Params
Without	**23.171**	118.013	0.2586	23.221M
only CLAM	22.942	<u>69.778</u>	0.2310	23.229M
only SLAM	22.938	71.004	<u>0.2289</u>	23.221M
CLAM+SLAM	<u>23.153</u>	**69.249**	**0.2110**	23.229M

reconstruction performance. Table VI shows the SR results of FastDiffSR with different sample schedules on the Toronto dataset. The linear sample schedule with timestep = 1000 can achieve higher visual quality, but the pixel-level metric PSNR is poor and time-consuming. When using the cosine sample schedule with timestep = 100, our FastDiffSR improves both the PSNR score and the average test time well. We also test different combinations of linear sample schedule and cosine sample schedule, and the model with the linear + 1.5 * cosine

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TABLE VI

EXPERIMENTAL RESULTS OF FASTDIFFSR WITH DIFFERENT SAMPLE SCHEDULES. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Sample schedule	Timestep	PSNR	FID	LPIPS	Average test time (s)
Linear	1000	22.295	**59.767**	**0.2100**	28.37774
Cosine	100	22.412	71.433	0.2230	7.29310
Linear + 0.5 * Cosine	20	22.875	97.114	0.2776	5.66144
Linear + 5.0 * Cosine		23.149	91.595	0.2462	5.71082
Linear + 1.5 * Cosine	10	**23.178**	109.820	0.2492	**5.19122**
	20	23.153	69.249	0.2110	5.27194
	50	22.838	91.596	0.2232	6.23589

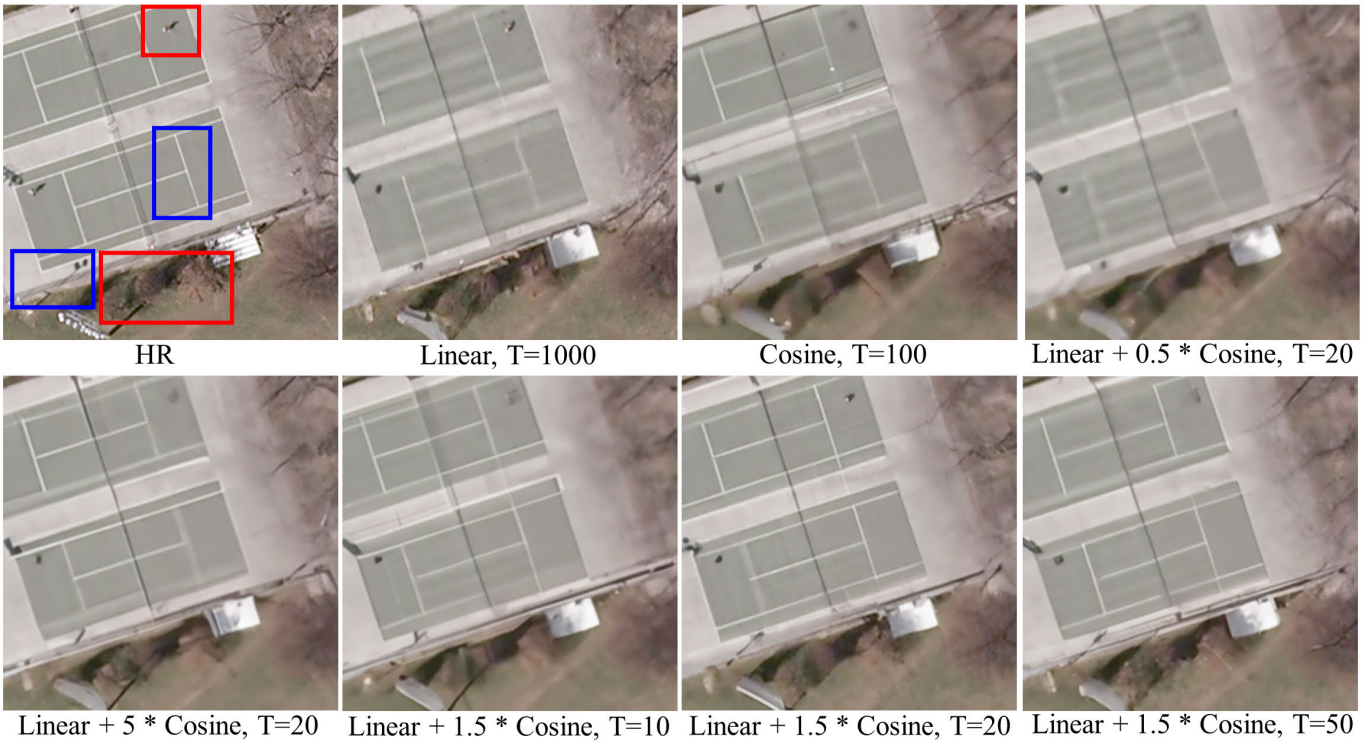


Fig. 10. Visual comparison of different sampling schedules in FastDiffSR.

TABLE VII

EXPERIMENTAL RESULTS OF DIFFERENT SR METHODS USING OUR FAST SAMPLING MODULE. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Method	Sample schedule	Timestep	PSNR	FID	LPIPS	Average test time (s)	Params
DDPM	Linear	1000	22.652	**94.789**	**0.2496**	19.31348	26.166M
SR3	Linear	2000	22.173	**56.022**	**0.1978**	45.61207	148.138M
SRDiff	Cosine	100	22.534	68.596	**0.2175**	4.17241	11.478M
FastDiffSR	Cosine	100	22.412	71.433	0.2230	7.29310	23.229M
DDPM	Linear + 1.5 * Cosine	20	**22.819**	105.573	0.2521	**3.73981**	26.166M
SR3			**22.419**	68.574	0.2052	**3.75157**	148.138M
SRDiff			**22.652**	**66.911**	0.2264	**3.40282**	11.478M
FastDiffSR			**23.153**	**69.249**	**0.2110**	**5.27194**	23.229M

sample strategy achieves the next best performance in all evaluation metrics. It can be found in Fig. 10 that the linear sample schedule can reconstruct the visual result closest to the HR image in the red region but cannot accurately generate the light and tennis court boundary in the blue region. On the contrary, the sampling strategy that combines the two methods can comprehensively balance the accuracy of the SR results and time efficiency. Therefore, our FastDiffSR applies the linear + 1.5 * cosine sample strategy with timestep = 20 as our final fast sampling module. For average test time, all results

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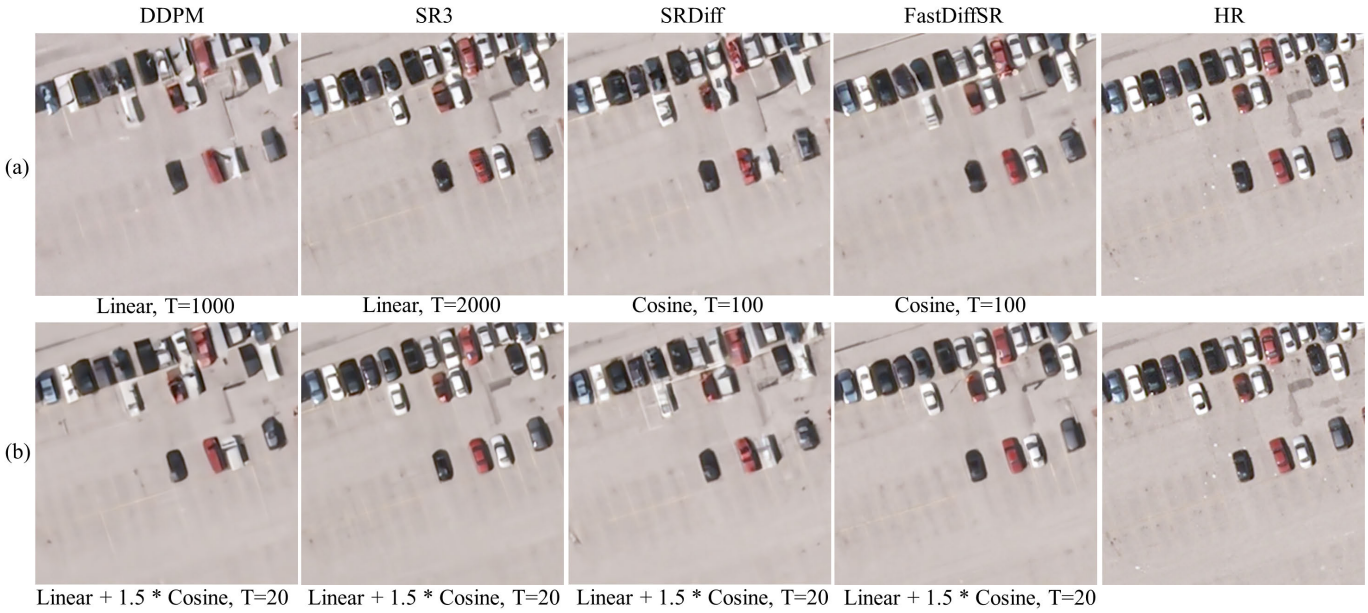


Fig. 11. Visual comparison of different SR methods using our fast sampling module. (a) Sampling without our fast sampling module. (b) Sampling with our fast sampling module.

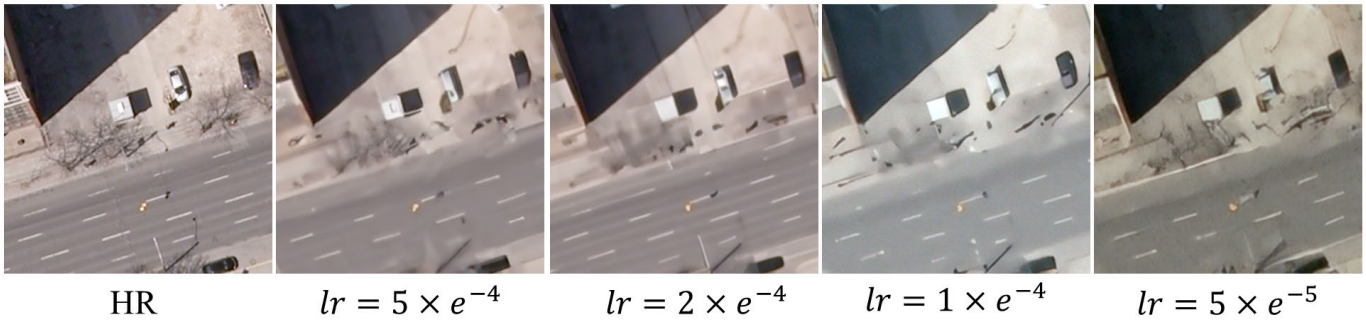


Fig. 12. Visual comparison of SR results for different learning rates.

are tested on an NVIDIA RTX3090 24-GB GPU with 256 × 256 HR images.

D. Effectiveness of the Fast Sampling Module

To verify the effectiveness of the proposed fast sampling module for diffusion-based methods, we employed DDPM [17], SR3 [19], SRDiff [43], and FastDiffSR methods for experimental analysis. Table VII reports the SR results of four models trained with our fast sampling module. It can be found that the linear sampling methods DDPM and SR3 have +0.167 and +0.246 dB in PSNR, respectively, and the running time per image is also significantly reduced. However, the disadvantage is that the FID and visual quality LPIPS metrics are slightly decreased. In addition, our fast sampling module is more helpful for the cosine sampling methods, which can reduce the number of sampling steps and running time while ensuring reconstruction accuracy. Fig. 11 also shows the qualitative comparison results of different methods trained with our fast sampling module. In general, our fast sampling module can better handle the generation problem of the diffusion model and obtain more accurate SR images consistent with the HR images.

TABLE VIII

EXPERIMENTAL RESULTS OF FASTDIFFSR WITH DIFFERENT LEARNING RATES. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Learning rate	PSNR	FID	LPIPS
5×10^{-4}	<u>22.924</u>	<u>107.645</u>	<u>0.2575</u>
2×10^{-4}	**23.153**	**69.249**	**0.2110**
1×10^{-4}	20.728	119.331	<u>0.2575</u>
5×10^{-5}	20.016	118.855	0.2876

E. Comparison of Different Learning Rates

To explore the impact of different learning rates on the performance of the proposed method, we conducted four sets of controlled experiments. As shown in Table VIII, the FastDiffSR with $\text{lr} = 2\text{e}^{-4}$ gets the best value on four metrics. Compared with $\text{lr} = 5\text{e}^{-5}$, the improvement reaches 3.137 dB in PSNR, 49.606 in FID, and 0.0766 in LPIPS. Fig. 12 shows the visual results of the FastDiffSR model with different learning rates on the Toronto dataset.

F. Model Complexity

To demonstrate the sampling efficiency of FastDiffSR, we select the Vaihingen dataset [49] for model complexity

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TABLE IX
MODEL COMPLEXITY ANALYSIS WITH SOTA SR MODELS ON THE VAIHINGEN IMAGE. BOLD REPRESENTS THE BEST RESULT AND UNDERLINE DENOTES THE SECOND BEST

Type	Method	Params	Memory (MB)	PSNR	FID	LPIPS	Inference time (s)
Non-Diffusion-based	Bicubic	/	/	33.287	255.397	0.6431	/
	SwinIR	<u>11.999M</u>	3298	33.096	364.212	0.4969	0.24194
	HSENet	**5.581M**	<u>2630</u>	<u>34.410</u>	231.011	0.5185	0.23759
	TransENet	37.607M	3980	32.746	279.478	0.5216	**0.12950**
	NDSRGAN	17.510M	2736	25.069	342.782	0.5651	<u>0.15195</u>
	HAT	25.821M	4948	34.329	275.856	0.5202	0.32875
Diffusion-based	DDPM	26.166M	10966	33.009	<u>220.483</u>	<u>0.2725</u>	26.06063
	TESR	105.128M	19866	31.449	378.862	0.4166	56.79351
	GDP_x_0	258.844M	20808	32.474	262.677	0.3864	53.82498
	EDiffSR	26.790M	**2207**	32.411	299.798	0.3164	7.53968
	Ours	23.229M	19186	**34.572**	**188.666**	**0.1415**	1.96339

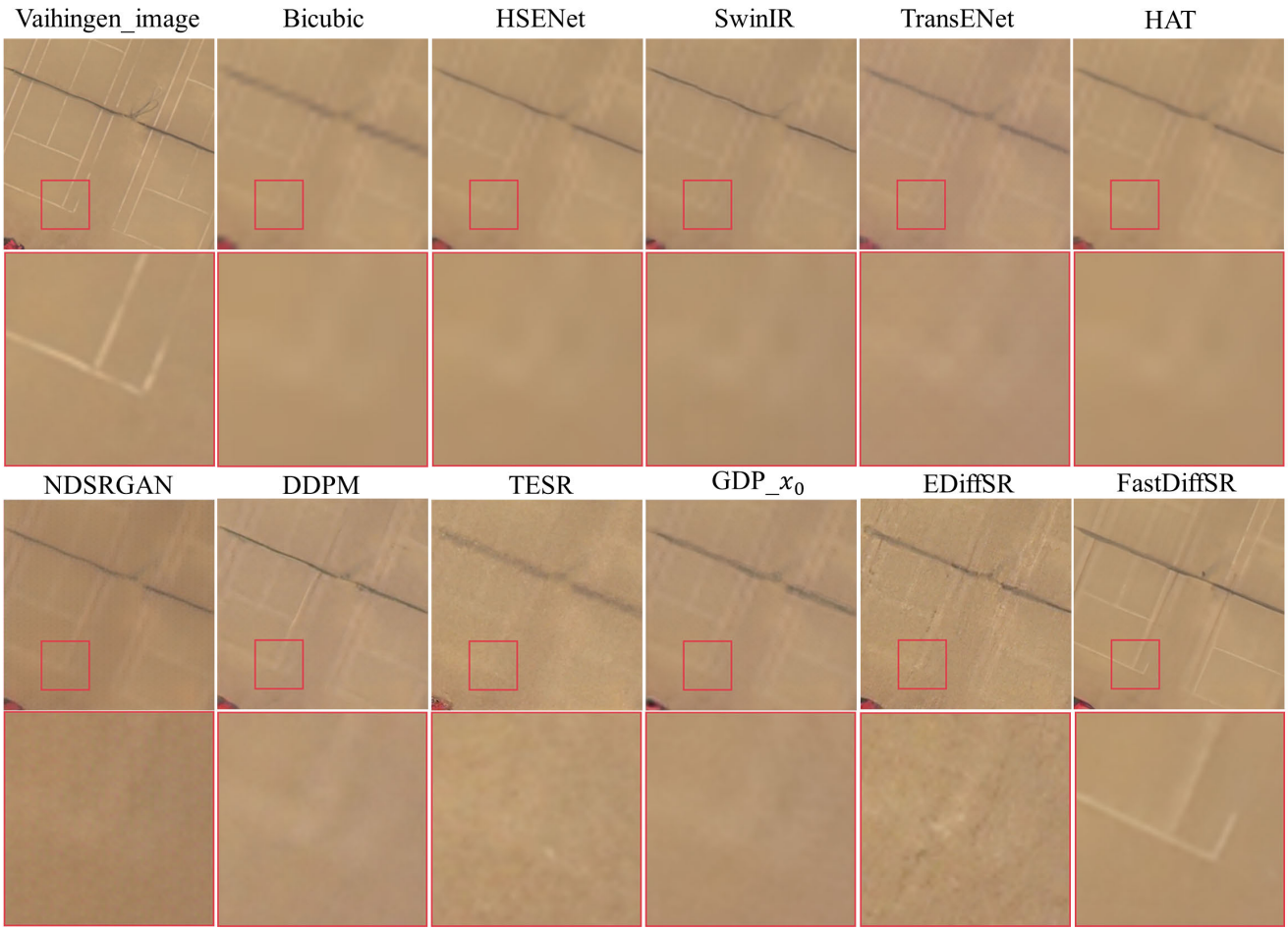


Fig. 13. ×8 SR results of different SOTA methods on Vaihingen image.

analysis. Similar to the Potsdam area, Vaihingen is a relatively small village. This dataset provides red (R), green (G), and near-infrared (NIR) three-band images, with a ground sampling distance of 0.09 m. We first perform bicubic downsampling of Vaihingen images (256 × 256) as input and then use trained ×8 scale SR models for corresponding SR reconstruction. Table IX reports the quantitative comparison of model complexities using six metrics, including model

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parameters, memory footprint, inference time, PSNR, FID, and LPIPS. Our proposed FastDiffSR significantly outperforms other methods in all reconstruction metrics. Although the inference time is lower than the nondiffusion-based method, it is 2.8–28 times higher than the Diffusion-based method. When compared with EDiffSR, the memory footprint of our method is not superior, which is a direction we need to improve in the future. Fig. 13 also lists the visualization results of different SOTA SR models. As can be seen, our method can still clearly recover the corners of the tennis court on an eight-time SR task. In comparison, only our method can recover more accurate textures while maintaining inference efficiency.

VII. Conclusion

We proposed a conditional diffusion model SR (FastDiffSR) method to generate high visual quality RSIs while achieving sampling efficiency. Compared with existing diffusion-based methods, we developed a new sampling technique to reduce the number of sampling steps of the diffusion model. Specifically, we combine a linear schedule and a cosine schedule to achieve a 20-step sampling process while ensuring sampling quality. To reduce the computational cost, we also introduce residual images and demonstrate that using channel attention and spatial attention can further improve the LPIPS score. Experiments on two benchmark datasets show that our method outperforms CNN-based, GAN-based, and Transformer-based methods in visual quality, and quantitative results outperform SOTA diffusion-based methods while having a smaller model size. In addition, FastDiffSR generates better SR results than other SOTA SR methods in some real-world scenes. More accurate SR results are obtained on Vaihingen data, and inference time is 2.8–28 times faster than the diffusion-based method, showing excellent generalization ability and time efficiency. In the future, we plan to extend our FastDiffSR to the more challenging undefined SR tasks of real-world RSIs to improve its real-world practical applications.

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